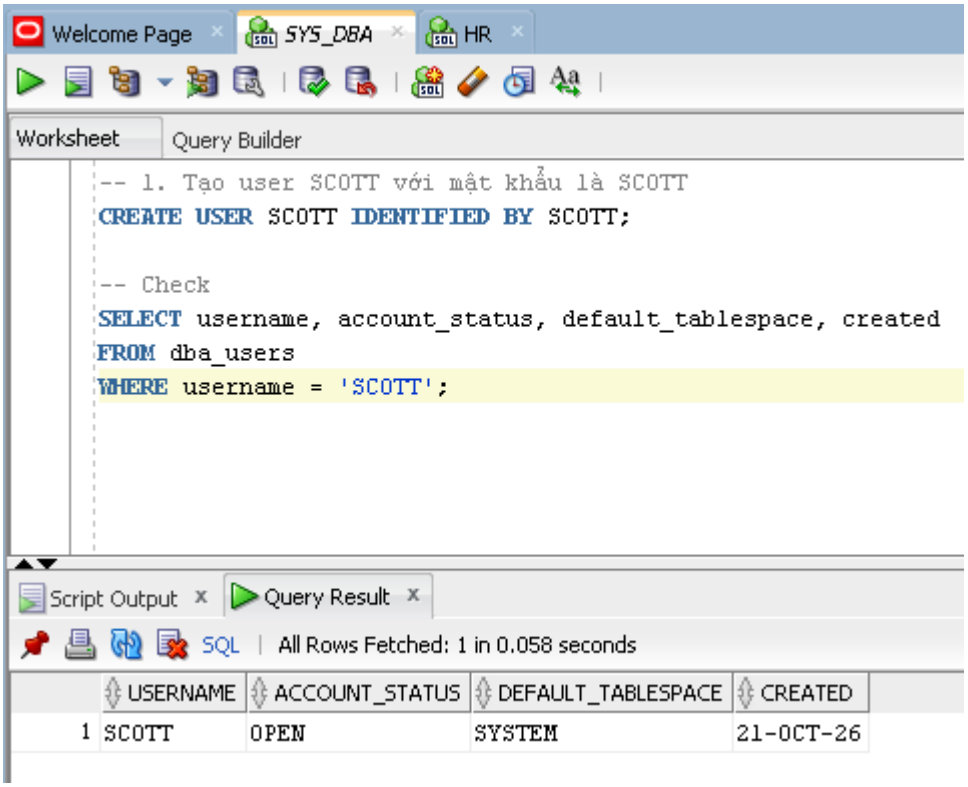
1. Create a user named SCOTT with the password as the same username.



2. List all users in the system.

```sql
-- 2. Liệt kê tất cả các user trong hệ thống từ view DBA_USERS
SELECT username, account_status, default_tablespace
FROM dba_users;
```

Script Output | Query Result

SQL | All Rows Fetched: 17 in 0.009 seconds

	USERNAME	ACCOUNT_STATUS	DEFAULT_TABLESPACE
1	SCOTT	OPEN	SYSTEM
2	SYS	OPEN	SYSTEM
3	SYSTEM	OPEN	SYSTEM
4	ANONYMOUS	OPEN	SYSAUX
5	HR	OPEN	USERS
6	APEX_PUBLIC_USER	LOCKED	SYSTEM
7	FLOWS_FILES	LOCKED	SYSAUX
8	APEX_040000	LOCKED	SYSAUX
9	OUTLN	EXPIRED & LOCKED	SYSTEM
10	DIP	EXPIRED & LOCKED	SYSTEM
11	ORACLE_OCM	EXPIRED & LOCKED	SYSTEM
12	XS$NULL	EXPIRED & LOCKED	SYSTEM
13	MDSYS	EXPIRED & LOCKED	SYSAUX
14	CTXSYS	EXPIRED & LOCKED	SYSAUX
15	DBSNMP	EXPIRED & LOCKED	SYSAUX
16	XDB	EXPIRED & LOCKED	SYSAUX
17	APPQOSSYS	EXPIRED & LOCKED	SYSAUX

3. Create an external user (operating system) and connect using that user.

```
--3
CREATE USER "OPS$EXTERNAL_USER1" IDENTIFIED EXTERNALLY;

GRANT CREATE SESSION TO "OPS$EXTERNAL_USER1";
```

Script Output | Query Result | Task completed in 0.058 seconds

```
User HR altered.


User SCOTT created.


User "OPS$EXTERNAL_USER1" created.


Grant succeeded.
```

TEST:

```
SELECT username, account_status, authentication_type
FROM dba_users
WHERE username = 'OPS$EXTERNAL_USER1';
```

Script Output | Query Result | SQL | All Rows Fetched: 1 in 0.006 seconds

	USERNAME	ACCOUNT_STATUS	AUTHENTICATION_TYPE
1	OPS$EXTERNAL_USER1	OPEN	EXTERNAL

4. Create a user SHIMON with the following options:

```
-- 4. Tạo user SHIMON
CREATE USER SHIMON IDENTIFIED BY SHIMON
DEFAULT TABLESPACE users
TEMPORARY TABLESPACE temp
QUOTA 10M ON users
PASSWORD EXPIRE
ACCOUNT LOCK;
```

Script Output | Query Result | Task completed in 0.04 seconds

```
User SHIMON created.
```

TEST:

```
--TEST:
SELECT username, account_status, default_tablespace, temporary_tablespace
FROM dba_users
WHERE username = 'SHIMON';
```

Script Output | Query Result | SQL | All Rows Fetched: 1 in 0.006 seconds

	USERNAME	ACCOUNT_STATUS	DEFAULT_TABLESPACE	TEMPORARY_TABLESPACE
1	SHIMON	EXPIRED & LOCKED	USERS	TEMP

5. Connect to the database using the SHIMON account

```
-- 5.
ALTER USER SHIMON IDENTIFIED BY SHIMON123;

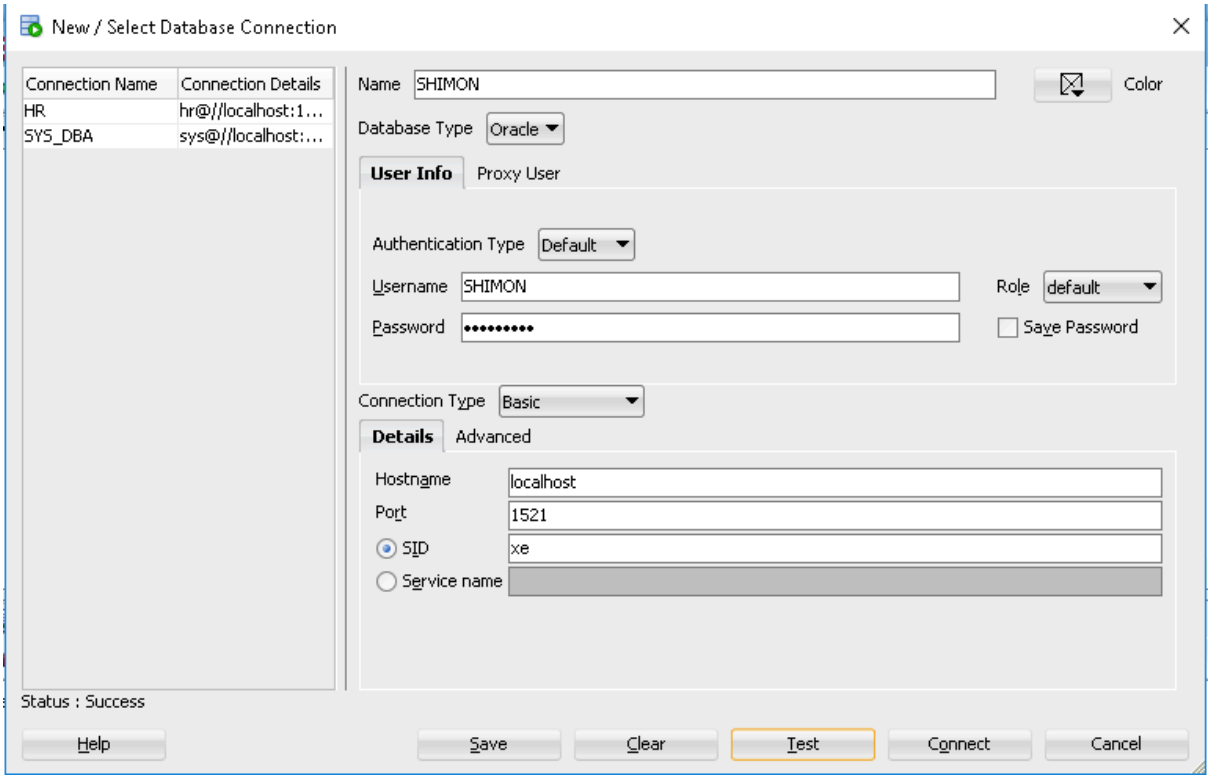
ALTER USER SHIMON ACCOUNT UNLOCK;
```

Script Output | Query Result | Task completed in 0.022 seconds

```
User SHIMON altered.


User SHIMON altered.
```

TEST KẾT NỐI:



6. Create a table ACCOUNT with two fields: acc_no (primary key) and balance.

```
-- 6. Tạo bảng ACCOUNT trong schema SHIMON
CREATE TABLE SHIMON.ACCOUNT (
    acc_no NUMBER PRIMARY KEY,
    balance NUMBER(15, 2)
);
```

Script Output | Query Result | Task completed in 0.119 seconds

```
Table SHIMON.ACCOUNT created.
```

TEST:

```
-- test
SELECT table_name, column_name, data_type, data_length
FROM dba_tab_columns
WHERE owner = 'SHIMON' AND table_name = 'ACCOUNT';
```

Script Output | Query Result | SQL | All Rows Fetched: 2 in 0.03 seconds

	TABLE_NAME	COLUMN_NAME	DATA_TYPE	DATA_LENGTH
1	ACCOUNT	ACC_NO	NUMBER	22
2	ACCOUNT	BALANCE	NUMBER	22

7. Create a role BASIC with the following privileges and grant this role to the user SHIMON:

```sql
--7.
--Tạo Role tên BASIC
CREATE ROLE BASIC;

--Cấp các quyền hệ thống cho Role BASIC
GRANT CREATE TABLE TO BASIC;
GRANT RESOURCE TO BASIC;
GRANT DROP ANY TABLE TO BASIC;

--Cấp quyền thao tác trên bảng HR.DEPARTMENTS cho Role BASIC
GRANT UPDATE ON HR.DEPARTMENTS TO BASIC;

--Gán Role BASIC cho user SHIMON
GRANT BASIC TO SHIMON;
```

Script Output | Query Result

Task completed in 0.021 seconds

```
Role BASIC created.


Grant succeeded.


Grant succeeded.


Grant succeeded.


Grant succeeded.


Grant succeeded.
```

TEST:

```sql
-- TEST
SELECT grantee, granted_role
FROM dba_role_privs
WHERE grantee = 'SHIMON';
```

Script Output | Query Result

SQL | All Rows Fetched: 1 in 0.006 seconds

	GRANTEE	GRANTED_ROLE
1	SHIMON	BASIC

8. Create a profile *dev_prof* with the following resource limits and change the profile of user SHIMON to the *dev_prof* profile:

```
--8.
--Bật tính năng giới hạn tài nguyên (RESOURCE_LIMIT) của hệ thống
ALTER SYSTEM SET RESOURCE_LIMIT = TRUE;

--Tạo Profile dev_prof với các thông số theo yêu cầu
CREATE PROFILE dev_prof LIMIT
    SESSIONS_PER_USER 2
    FAILED_LOGIN_ATTEMPTS 5
    PASSWORD_REUSE_MAX 4;

--Gán Profile dev_prof cho user SHIMON
ALTER USER SHIMON PROFILE dev_prof;\
```

Script Output | Query Result

Task completed in 0.044 seconds

```
System SET altered.


Profile DEV_PROF created.


User SHIMON altered.
```

TEST:

```
-- TEST
SELECT username, profile
FROM dba_users
WHERE username = 'SHIMON';
```

Script Output | Query Result

SQL | All Rows Fetched: 1 in 0.006 seconds

	USERNAME	PROFILE
1	SHIMON	DEV_PROF